

Probability and Stochastic Processes

Q1: If the probability density function (PDF) of a random variable X is given by $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, what type of distribution does X have?

- a) Poisson
- b) Exponential
- c) Gaussian
- d) Uniform

Answer: b) Exponential

2. Poisson Process

Q2: In a Poisson process with a rate parameter $\lambda = 3$ events per minute, what is the probability of exactly 2 events occurring in a 1-minute interval?

- a) $\frac{9e^{-3}}{2}$
- b) $\frac{6e^{-3}}{2}$
- c) $\frac{9e^{-2}}{2}$
- d) e^{-3}

Answer: a) $\frac{9e^{-3}}{2}$

3. Sampling Theory

Q3: A signal is band-limited to 5 kHz. What is the minimum sampling frequency required to avoid aliasing?

- a) 5 kHz
- b) 10 kHz
- c) 15 kHz
- d) 20 kHz

Answer: b) 10 kHz

4. Pulse Code Modulation (PCM)

Q4: In a PCM system with 8-bit quantization, how many quantization levels are available?

- a) 8
- b) 16
- c) 128
- d) 256

Answer: d) 256

5. Delta Modulation

Q5: Which of the following issues is addressed by Adaptive Delta Modulation but not by standard Delta Modulation?

- a) Slope overload distortion
- b) Quantization noise
- c) Low bandwidth utilization
- d) Inadequate sampling

Answer: a) Slope overload distortion